

On the Structural Plastica

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Abstract

The Structural Plastica Theory (SPT) is based on the concept of spacetime as a deformable, elastic solid called the “structural plastica.” In viewing spacetime through this lens, we can reproduce physical phenomena like Newtonian gravity without invoking heavy concepts like those that make up differential geometry.

SPT begins with the idea of the plastica as a coordinate system embedded within a container. Relating accelerations inside both spaces gives a phenomena called Lavelle deflection, which is the fictitious force enacted on objects by the curvature of the plastica. We’ll then relate the plastica’s curvature to mass via a Lagrangian action and plug in a particle’s mass distribution to get Newtonian gravity from Lavelle deflection.

Note that while SPT is a theory of gravity, it is not built to hold up to the Theory of Relativity; instead, it is a “toy model” of reality based on the premise of the plastica.

Before we begin, it’s important to note that we do **not** have a metric in SPT for raising and lowering indices. Index placement is purely notational for Einstein summation and does not indicate covariance or contravariance; instead, it simply indicates a summation when one is raised and the other is lowered.

Section 1: Plastica Kinematics

1.1 Lavelle’s Equation

To start, we’ll define the map from plastica coordinates to container coordinates as $\mathbf{C}(\mathbf{x})$:

$$\mathbf{C}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^4$$

where $\mathbf{C}$ represents container coordinates and $\mathbf{x}$ represents plastica coordinates. Both are 4-vectors, as SPT’s spacetime includes three spatial and one temporal direction.

By the standard identity for coordinate maps, $d\mathbf{x} = \partial\mathbf{x}/\partial\mathbf{C} d\mathbf{C}$, so:

$$dx_\mu = \frac{\partial x_\mu}{\partial C_i} dC^i$$

Naming the “forward” Jacobian $\frac{\partial x_\mu}{\partial C_i} = \Lambda_{\mu i}$:

$$dx_\mu = \Lambda_{\mu i} dC^i$$

The above equation will be referred to as Lavelle’s equation. Notice the matrix structure of this result; along any axis μ , the distance measure does not depend on dC^μ alone, but instead the sum of contributions from other directions weighted by $\Lambda_{\mu i}$. Dividing through by dC_4 :

$$\dot{x}_\mu = \Lambda_{\mu i} \frac{dC^i}{dC_4} = \Lambda_{\mu i} \dot{C}^i$$

with dots w.r.t C_4 , the time dimension. Differentiating once more and using the product rule:

$$\ddot{x}_\mu = \frac{d}{dC_4} (\Lambda_{\mu i} \dot{C}^i) = \frac{d\Lambda_{\mu i}}{dC_4} \dot{C}^i + \Lambda_{\mu i} \ddot{C}^i$$

Using the chain rule on $d\Lambda_{\mu i}/dC_4$:

$$\frac{d\Lambda_{\mu i}}{dC_4} = \frac{\partial \Lambda_{\mu i}}{\partial C_j} \frac{dC^j}{dC_4} = \frac{\partial \Lambda_{\mu i}}{\partial C_j} \dot{C}^j \implies \ddot{x}_\mu = \frac{\partial \Lambda_{\mu i}}{\partial C_j} \dot{C}^i \dot{C}^j + \Lambda_{\mu i} \ddot{C}^i$$

Converting the $\dot{C}^i$ and $\dot{C}^j$ to plastica velocities:

$$\dot{C}^i = \frac{\partial C^i}{\partial x^k} \frac{dx_k}{dC_4} = \frac{\partial C^i}{\partial x^k} \dot{x}_k$$

Naming the “backward” Jacobian $\frac{\partial C^i}{\partial x^k}$ to Θ^{ik} :

$$\dot{C}^i = \Theta^{ik} \dot{x}_k \implies \ddot{x}_\mu = \frac{\partial \Lambda_{\mu i}}{\partial C_j} \Theta^{ik} \Theta^{jl} \dot{x}_k \dot{x}_l + \Lambda_{\mu i} \ddot{C}^i$$

Note that Θ and Λ are inverses, so:

$$\begin{aligned} \Lambda_{ab} \Theta^{bc} &= \delta_a^c \implies \frac{\partial}{\partial x^d} (\delta_a^c) = 0 = \frac{\partial}{\partial x^d} (\Lambda_{ab} \Theta^{bc}) = \frac{\partial \Lambda_{ab}}{\partial x^d} \Theta^{bc} + \Lambda_{ab} \frac{\partial \Theta^{bc}}{\partial x^d} \\ \frac{\partial \Lambda_{ab}}{\partial x^d} &= \frac{\partial \Lambda_{ab}}{\partial C_e} \frac{\partial C^e}{\partial x^d} = \frac{\partial \Lambda_{ab}}{\partial C_e} \Theta^{ed} \implies 0 = \frac{\partial \Lambda_{ab}}{\partial C_e} \Theta^{ed} \Theta^{bc} + \Lambda_{ab} \frac{\partial \Theta^{bc}}{\partial x^d} \\ &\implies \frac{\partial \Lambda_{ab}}{\partial C_e} \Theta^{bc} \Theta^{ed} = -\Lambda_{ab} \frac{\partial \Theta^{bc}}{\partial x^d} \end{aligned}$$

Plugging into $\ddot{x}_\mu$, with $a = \mu, b = i, c = k, d = l$:

$$\ddot{x}_\mu = -\Lambda_{\mu i} \frac{\partial \Theta^{ik}}{\partial x^l} \dot{x}_k \dot{x}_l + \Lambda_{\mu i} \ddot{C}^i$$

Naming the “curvature” tensor $\frac{\partial \Theta^{ik}}{\partial x^l}$ to Ω^{ikl} :

$$\ddot{x}_\mu = -\Lambda_{\mu i} \Omega^{ikl} \dot{x}_k \dot{x}_l + \Lambda_{\mu i} \ddot{C}^i$$

Notice that only one term requires an external container force $\ddot{C}^i$; that term bends incoming forces using the forward Jacobian. The other term appears due to the second-order change of the plastica and only depends on velocity. Dropping the external container force to work with free particles, we get Lavelle deflection:

$$\ddot{x}_\mu = -\Lambda_{\mu i} \Omega^{ikl} \dot{x}_k \dot{x}_l$$

1.2 Perturbations To First Order

Before we move into gravity, it’s important that we set the stage first. To do so, we’ll be deriving Lavelle deflection for a map given by:

$$C_\mu = x_\mu + \epsilon \alpha_\mu$$

where ϵ is some small parameter and α_i is the perturbation structure. We’ll be tracking our result to first order in ϵ , discarding any result higher than second order in ϵ .

Starting with the backward Jacobian:

$$\Theta_{ij} = \frac{\partial x_i}{\partial x_j} + \epsilon \frac{\partial \alpha_i}{\partial x_j} = \delta_{ij} + \epsilon \frac{\partial \alpha_i}{\partial x_j}$$

which gives us the forward Jacobian to first order in ϵ :

$$\Rightarrow \Lambda_{ij} = \Theta_{ij}^{-1} = \delta_{ij} - \epsilon \frac{\partial \alpha_i}{\partial x_j} + O(\epsilon^2)$$

Next, the curvature tensor:

$$\Omega_{ijk} = \frac{\partial \Theta_{ij}}{\partial x_k} = \frac{\partial \delta_{ij}}{\partial x_k} + \epsilon \frac{\partial^2 \alpha_i}{\partial x_j \partial x_k} = \epsilon \frac{\partial^2 \alpha_i}{\partial x_j \partial x_k}$$

Now Lavelle deflection:

$$\begin{aligned} \ddot{x}_\mu &= -\Lambda_{\mu i} \Omega^{ikl} \dot{x}_k \dot{x}_l = - \left(\delta_{\mu i} - \epsilon \frac{\partial \alpha_\mu}{\partial x_i} \right) \left(\epsilon \frac{\partial^2 \alpha^i}{\partial x^k \partial x^l} \right) \dot{x}_k \dot{x}_l \\ &= -\epsilon \frac{\partial^2 \alpha_\mu}{\partial x^k \partial x^l} \dot{x}_k \dot{x}_l \end{aligned}$$

Moving the $\partial x^k \partial x^l$ indices down and the $\dot{x}_k \dot{x}_l$ indices up, we get our final result; linearized Lavelle deflection equation:

$$\ddot{x}_\mu = -\epsilon \frac{\partial^2 \alpha_\mu}{\partial x_k \partial x_l} \dot{x}^k \dot{x}^l$$

Section 2: Dynamics

2.1 The Sourcing Equation

To source the plastica, we'll be using a Lagrangian action and the Euler-Lagrange equations. Our action goes as:

$$S = \int \mathcal{L} d^3x = \int |\det \Lambda| \Omega^{\nu mn} \Omega_{\nu mn} - \omega^\nu \frac{\partial m}{\partial x_\nu} d^3x \quad \omega^\nu = \frac{\partial^2 C^\nu}{\partial x^k \partial x_k} = \Omega^{\nu k}_k$$

where ν runs over spatial indices, m, n run over all indices, and m is our mass density. The first term is the squared norm of the curvature, weighted by volume expansion; the second term is a sourcing term generated by mass bound to the vector Laplacian of $\mathbf{C}$. We'll refer to the first side of $\mathcal{L}$ as $\mathcal{L}_1$ and the second side as $\mathcal{L}_2$. Notice that this integral is run over three dimensions, not four. This will be important later, as we'll find that a full 4D equation would overdetermine the system.

The Euler-Lagrange equations give:

$$\frac{\partial \mathcal{L}}{\partial C^\eta} - \frac{\partial}{\partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Theta^{\eta\rho}} + \frac{\partial^2}{\partial x_\sigma \partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Omega^{\eta\rho\sigma}} = 0$$

where η runs over the first three indices. The first term is simple, as no term contains $\mathbf{C}$ directly:

$$\frac{\partial \mathcal{L}}{\partial C^\eta} = 0$$

For the second, we'll use Jacobi's formula on $|\det \Lambda| = |\det \Theta^{-1}|$:

$$\begin{aligned} \frac{\partial \mathcal{L}_1}{\partial \Theta^{\eta\rho}} &= \frac{\partial}{\partial \Theta^{\eta\rho}} (|\det \Lambda|) \Omega^{\nu mn} \Omega_{\nu mn} = -|\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} \\ &\Rightarrow -\frac{\partial}{\partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Theta^{\eta\rho}} = \frac{\partial}{\partial x_\rho} (|\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn}) \\ &+ \frac{\partial}{\partial x_\rho} (|\det \Lambda|) \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} + |\det \Lambda| \frac{\partial \Lambda^{\eta\rho}}{\partial x_\rho} \Omega^{\nu mn} \Omega_{\nu mn} \\ &= +|\det \Lambda| \Lambda^{\eta\rho} \frac{\partial \Omega^{\nu mn}}{\partial x_\rho} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \frac{\partial \Omega_{\nu mn}}{\partial x_\rho} \end{aligned}$$

We'll name the tensor $\partial \Omega_{\nu mn} / \partial x_\rho = \partial^3 C_\nu / \partial x_m \partial x_n \partial x_\rho = \Phi_{\nu mn\rho}$ and use the chain rule on both $|\det \Lambda|$ and $\Lambda^{\eta\rho}$:

$$\begin{aligned} \frac{\partial}{\partial x_\rho} (|\det \Lambda|) &= \frac{\partial}{\partial \Theta^{ab}} (|\det \Lambda|) \frac{\partial \Theta^{ab}}{\partial x_\rho} = -|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \\ \frac{\partial \Lambda^{\eta\rho}}{\partial x_\rho} &= \frac{\partial \Lambda^{\eta\rho}}{\partial \Theta^{cd}} \frac{\partial \Theta^{cd}}{\partial x_\rho} = -\Lambda^{c\eta} \Lambda^{\rho d} \Omega_{cd\rho} \end{aligned}$$

$$\begin{aligned} \Rightarrow -\frac{\partial}{\partial x_\rho} \frac{\partial \mathcal{L}_1}{\partial \Theta^{\eta\rho}} &= +|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{c\eta} \Lambda^{\rho d} \Omega_{cd\rho} \Omega^{\nu mn} \Omega_{\nu mn} \\ &+ |\det \Lambda| \Lambda^{\eta\rho} \frac{\partial \Omega^{\nu mn}}{\partial x_\rho} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \frac{\partial \Omega_{\nu mn}}{\partial x_\rho} \end{aligned}$$

Noticing that we can combine the last two terms and using Φ :

$$\begin{aligned} &+ |\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{c\eta} \Lambda^{\rho d} \Omega_{cd\rho} \Omega^{\nu mn} \Omega_{\nu mn} \\ &= +2|\det \Lambda| \Lambda^{\eta\rho} \Phi_{\nu mn\rho} \Omega^{\nu mn} \end{aligned}$$

Next, we'll take the third term for the first side of $\mathcal{L}$:

$$\begin{aligned} \frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} &= \frac{\partial}{\partial \Omega^{\eta\rho\sigma}} (\Omega^{\nu mn} \Omega_{\nu mn}) |\det \Lambda| = |\det \Lambda| \left(\frac{\partial \Omega^{\nu mn}}{\partial \Omega^{\eta\rho\sigma}} \Omega_{\nu mn} + \Omega^{\nu mn} \frac{\partial \Omega_{\nu mn}}{\partial \Omega^{\eta\rho\sigma}} \right) \\ &= |\det \Lambda| (\delta^{\eta\nu} \delta^{\rho m} \delta^{\sigma n} \Omega_{\nu mn} + \Omega^{\nu mn} \delta_\nu^\eta \delta_m^\rho \delta_n^\sigma) = |\det \Lambda| (\Omega^{\eta\rho\sigma} + \Omega^{\eta\rho\sigma}) = 2|\det \Lambda| \Omega^{\eta\rho\sigma} \\ &\Rightarrow \frac{\partial^2}{\partial x_\sigma \partial x_\rho} \frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} = 2 \frac{\partial}{\partial x_\sigma} \left(\frac{\partial}{\partial x_\rho} (|\det \Lambda| \Omega^{\eta\rho\sigma}) \right) \\ &= 2 \frac{\partial}{\partial x_\sigma} \left(\frac{\partial}{\partial x_\rho} (|\det \Lambda|) \Omega^{\eta\rho\sigma} + |\det \Lambda| \frac{\partial \Omega^{\eta\rho\sigma}}{\partial x_\rho} \right) \end{aligned}$$

By previous results:

$$\begin{aligned} &\frac{\partial}{\partial x_\rho} (|\det \Lambda|) = -|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \\ &\Rightarrow \frac{\partial^2}{\partial x_\sigma \partial x_\rho} \frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} = 2 \frac{\partial}{\partial x_\sigma} (-|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} + |\det \Lambda| \Phi^{\eta\rho\sigma}_\rho) \\ &= -2 \frac{\partial}{\partial x_\sigma} (|\det \Lambda|) \Lambda^{ab} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} - 2|\det \Lambda| \frac{\partial \Lambda^{ab}}{\partial x_\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} - 2|\det \Lambda| \Lambda^{ab} \frac{\partial \Omega_{ab\rho}}{\partial x_\sigma} \Omega^{\eta\rho\sigma} \\ &= -2|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \frac{\partial \Omega^{\eta\rho\sigma}}{\partial x_\sigma} + 2 \frac{\partial}{\partial x_\sigma} (|\det \Lambda|) \Phi^{\eta\rho\sigma}_\rho + 2|\det \Lambda| \frac{\partial \Phi^{\eta\rho\sigma}_\rho}{\partial x_\sigma} \end{aligned}$$

Naming the tensor $\partial^4 C^\eta / \partial x^\rho \partial x^\sigma \partial x_\rho \partial x_\sigma = \Psi^{\eta\sigma}$ and expanding derivatives:

$$\begin{aligned} &+ 2|\det \Lambda| \Lambda^{cd} \Omega_{cd\sigma} \Lambda^{ab} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} + 2|\det \Lambda| \Lambda^{ca} \Lambda^{bd} \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} - 2|\det \Lambda| \Lambda^{ab} \Phi_{ab\rho\sigma} \Omega^{\eta\rho\sigma} \\ &= -2|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Phi^{\eta\rho\sigma}_\sigma - 2|\det \Lambda| \Lambda^{cd} \Omega_{cd\sigma} \Phi^{\eta\rho\sigma}_\rho + 2|\det \Lambda| \Psi^\eta \end{aligned}$$

Finally, the third term for the second side of $\mathcal{L}$:

$$\begin{aligned} \frac{\partial \mathcal{L}_2}{\partial \Omega^{\eta\rho\sigma}} &= -\frac{\partial m}{\partial x_\nu} \frac{\partial^2 \omega^\nu}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial m}{\partial x_\nu} \frac{\partial^2 \Omega^{\nu k}}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial m}{\partial x_\nu} \delta^{\eta\nu} \delta^{\rho k} \delta_k^\sigma \\ &\Rightarrow \frac{\partial^2}{\partial x_\sigma \partial x_\rho} \frac{\partial \mathcal{L}_2}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial^2}{\partial x_\sigma \partial x_\rho} \left(\frac{\partial m}{\partial x_\nu} \delta^{\eta\nu} \delta^{\rho k} \delta_k^\sigma \right) = -\frac{\partial^2}{\partial x_\sigma \partial x_\rho} \left(\frac{\partial m}{\partial x_\nu} \right) \delta^{\eta\nu} \delta^{\rho k} \delta_k^\sigma \end{aligned}$$

$$= -\frac{\partial^3 m}{\partial x_\sigma \partial x_\rho \partial x_\nu} \delta^{\eta\nu} \delta^{\rho k} \delta_k^\sigma = -\frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta}$$

Bringing all the terms together:

$$\begin{aligned} & + |\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{c\eta} \Lambda^{\rho d} \Omega_{cd\rho} \Omega^{\nu mn} \Omega_{\nu mn} + 2|\det \Lambda| \Lambda^{\eta\rho} \Phi_{\nu mn\rho} \Omega^{\nu mn} \\ & + 2|\det \Lambda| \Lambda^{cd} \Omega_{cd\sigma} \Lambda^{ab} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} + 2|\det \Lambda| \Lambda^{ca} \Lambda^{bd} \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} - 2|\det \Lambda| \Lambda^{ab} \Phi_{ab\rho\sigma} \Omega^{\eta\rho\sigma} \\ 0 = & -2|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho} \Phi^{\eta\rho\sigma}{}_\sigma - 2|\det \Lambda| \Lambda^{cd} \Omega_{cd\sigma} \Phi^{\eta\rho\sigma}{}_\rho + 2|\det \Lambda| \Psi^\eta \\ & - \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} \end{aligned}$$

Factoring, moving the source term, collapsing summations, and using $\mathcal{L}_1$'s definition:

$$\begin{aligned} \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = & \mathcal{L}_1 \Omega_{ab\rho} (\Lambda^{ab} \Lambda^{\eta\rho} + \Lambda^{a\eta} \Lambda^{\rho b}) + 2|\det \Lambda| \left(\begin{aligned} & + \Lambda^{\eta\rho} \Omega^{\nu mn} \Phi_{\nu mn\rho} \\ & + \Lambda^{cd} \Lambda^{ab} \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} \\ & + \Lambda^{ca} \Lambda^{bd} \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} \\ & - \Lambda^{ab} \Omega^{\eta\rho\sigma} \Phi_{ab\rho\sigma} \\ & - \Lambda^{ab} \Omega_{ab\rho} \Phi^{\eta\rho\sigma}{}_\sigma \\ & - \Lambda^{cd} \Omega_{cd\sigma} \Phi^{\eta\rho\sigma}{}_\rho \end{aligned} \right) + 2|\det \Lambda| \Psi^\eta \\ \frac{1}{2} \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = & |\det \Lambda| \Psi^\eta + |\det \Lambda| \left(\begin{aligned} & - \Lambda^{ab} \Omega^{\eta\rho\sigma} \Phi_{ab\rho\sigma} - 2\Lambda^{ab} \Omega_{ab\rho} \Phi^{\eta\rho\sigma}{}_\sigma \\ & + \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} (\Lambda^{ab} \Lambda^{cd} + \Lambda^{ca} \Lambda^{bd}) \end{aligned} \right) + \frac{1}{2} \mathcal{L}_1 \Omega_{ab\rho} (\Lambda^{ab} \Lambda^{\eta\rho} + \Lambda^{a\eta} \Lambda^{\rho b}) \end{aligned}$$

This is the sourcing equation of the plastica in its full form.

2.2 Analyzing the Sourcing Equation

Before we begin analyzing, let's discuss what adding and subtracting a term means in the sourcing equation. For a fixed LHS, adding a positive term means there is more effect for the same source; thus, added terms make the plastica more reactive. Subtracting a term means there is less effect for the same source; thus, subtracted terms make the plastica stiffer.

A few semantic terms I will be using should also be discussed beforehand; “bound” quantities are the factors of a product, and thus follow a multiplicative relationship. For a fixed product, one must increase when the other decreases, and both must be nonzero. “Stretch” refers to the physical stretching of the plastica, represented by Λ and Θ .

If we assume the ansatz $C_i = x_i + \epsilon \alpha_i$, we can track the orders of each sourcing equation term as well; we'll do so by assigning each building block an order:

$$\Lambda_{ij} = \delta_{ij} - \epsilon \frac{\partial \alpha_i}{\partial x_j} \sim O(1) \quad \Omega_{ijk} = \epsilon \frac{\partial^2 \alpha_i}{\partial x_j \partial x_k} \sim O(\epsilon) \quad \mathcal{L}_1 = |\det \Lambda| \Omega^{\nu mn} \Omega_{\nu mn} \sim O(\epsilon^2)$$

Φ and Ψ are $O(\epsilon)$ too, as they're derivatives of Ω .

Starting with the sourcing equation's first term:

$$|\det \Lambda| \Psi^\eta = \left| \det \frac{\partial \mathbf{x}}{\partial \mathbf{C}} \right| \frac{\partial^4 C^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu}$$

This term is a Λ - Ψ product, so it's $O(\epsilon)$. It's a biharmonic operator on C^η (double Laplacian) weighted by volume expansion in the plastica. Under the perturbation ansatz, this term goes as:

$$|\det \Lambda| \Psi^\eta = \epsilon \left| 1 - \epsilon \frac{\partial \alpha_i}{\partial x^i} \right| \frac{\partial^4 \alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu} = \epsilon \frac{\partial^4 \alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu} + O(\epsilon^2)$$

Next, the second term:

$$\begin{aligned} & |\det \Lambda| \left(\begin{aligned} & + \Lambda^{\eta\rho} \Omega^{\nu mn} \Phi_{\nu mn\rho} \\ & - \Lambda^{ab} \Omega^{\eta\rho\sigma} \Phi_{ab\rho\sigma} - 2\Lambda^{ab} \Omega_{ab\rho} \Phi^{\eta\rho\sigma}{}_\sigma \\ & + \Omega_{cd\sigma} \Omega_{ab\rho} \Omega^{\eta\rho\sigma} (\Lambda^{ab} \Lambda^{cd} + \Lambda^{ca} \Lambda^{bd}) \end{aligned} \right) \\ &= \left| \det \frac{\partial \mathbf{x}}{\partial \mathbf{C}} \right| \left(\begin{aligned} & + \frac{\partial x^\eta}{\partial C_\rho} \frac{\partial^2 C^\nu}{\partial x_m \partial x_n} \frac{\partial^3 C_\nu}{\partial x^m \partial x^n \partial x_\rho} \\ & - \frac{\partial x^a}{\partial C_b} \frac{\partial^2 C^\eta}{\partial x_\rho \partial x_\sigma} \frac{\partial^3 C_a}{\partial x^b \partial x_\rho \partial x_\sigma} - 2 \frac{\partial x^a}{\partial C_b} \frac{\partial^2 C_a}{\partial x^b \partial x_\rho} \frac{\partial^3 C^\eta}{\partial x_\rho \partial x_\sigma \partial x^\sigma} \\ & + \frac{\partial^2 C_c}{\partial x_d \partial x_\sigma} \frac{\partial^2 C_a}{\partial x_b \partial x_\rho} \frac{\partial^2 C^\eta}{\partial x_\rho \partial x_\sigma} \left(\frac{\partial x^a}{\partial C_b} \frac{\partial x^c}{\partial C_d} + \frac{\partial x^c}{\partial C_a} \frac{\partial x^d}{\partial C_b} \right) \end{aligned} \right) \end{aligned}$$

The first piece is Λ - Λ - Ω - Φ , which is $O(\epsilon^2)$, and binds the plastica's stretch, curvature, and change of curvature together. The second piece is Λ - Λ - Ω - Φ , giving $O(\epsilon^2)$ as well, mirroring the first piece. The third piece is Λ - Ω - Ω - Ω - Λ , $O(\epsilon^3)$, and binds curvature to stretch. At most, this term is cubic in curvature and quadratic in stretch; these terms are also already $O(\epsilon^2)$ or $O(\epsilon^3)$, so they have no linearized equivalent.

Finally, the third term:

$$\begin{aligned} & \frac{1}{2} \mathcal{L}_1 \Omega_{ab\rho} (\Lambda^{ab} \Lambda^{\eta\rho} + \Lambda^{a\eta} \Lambda^{\rho b}) \sim O(\epsilon^3) \\ &= \frac{1}{2} \left| \det \frac{\partial \mathbf{x}}{\partial \mathbf{C}} \right| \frac{\partial^2 C^\nu}{\partial x_m \partial x_n} \frac{\partial^2 C_\nu}{\partial x^m \partial x^n} \frac{\partial^2 C_a}{\partial x_b \partial x_\rho} \left(\frac{\partial x^a}{\partial C_b} \frac{\partial x^\eta}{\partial C_\rho} + \frac{\partial x^a}{\partial C_\rho} \frac{\partial x^\eta}{\partial C_b} \right) \end{aligned}$$

This term is $\mathcal{L}_1$ - Ω - Λ - Λ , which is $O(\epsilon^3)$. The Lagrangian density reappears here, binding volume expansion, curvature, and stretch together in a single term. Because this has no linearized equivalent, the full linearized sourcing equation comes out to:

$$\begin{aligned} & \frac{1}{2} \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = \epsilon \frac{\partial^4 \alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\sigma \partial x_\sigma} + O(\epsilon^2) \\ & \implies \frac{1}{2} \frac{\partial m}{\partial x^\eta} = \epsilon \frac{\partial^2 \alpha^\eta}{\partial x^\nu \partial x_\nu} = \Omega^{\eta\sigma}{}_\sigma \end{aligned}$$

Section 3: Gravity

3.1 Building Gravity

Take a point source for m , weighted quadratically by x_4 :

$$m = -\kappa \epsilon x_4^2 \delta^3(\mathbf{x}) \implies \frac{1}{2} \frac{\partial m}{\partial x^\eta} = -\frac{1}{2} \kappa \epsilon x_4^2 \cdot \frac{\partial \delta^3}{\partial x^\eta}$$

Thus, α is the Green's function of the RHS:

$$\frac{\partial^3}{\partial x^k \partial x_k \partial x^\eta} \left(-\frac{1}{4\pi |\mathbf{r}|} \right) = \frac{\partial \delta^3}{\partial x^\eta} \implies \alpha^\eta = -\frac{1}{8\pi} \kappa \epsilon x_4^2 \frac{\partial}{\partial x^\eta} \left(\frac{1}{|\mathbf{r}|} \right)$$

where $\mathbf{r} = (x_1, x_2, x_3)$. The standard result for $\partial/\partial x^\eta (1/|\mathbf{r}|^n)$ is $-n x^\eta / |\mathbf{r}|^{n+2}$, so:

$$\alpha^\eta = \frac{1}{8\pi} \kappa \epsilon x_4^2 \frac{x^\eta}{|\mathbf{r}|^3} \implies \alpha_\nu = \epsilon \kappa \frac{x_\nu}{|\mathbf{r}|^3}$$

with κ now equal to $8\pi k$.

Differentiating once for spatial index η :

$$\frac{\partial \alpha_\nu}{\partial x_\eta} = \kappa x_4^2 \frac{\partial}{\partial x_\eta} \left(\frac{x_\nu}{|\mathbf{r}|^3} \right) = \kappa x_4^2 \left(\frac{\partial x_\nu}{\partial x_\eta} \frac{1}{|\mathbf{r}|^3} + x_\nu \frac{\partial}{\partial x_\eta} \left(\frac{1}{|\mathbf{r}|^3} \right) \right)$$

The standard result for $\partial/\partial x_\eta (1/|\mathbf{r}|^n)$ is $-n x_\eta / |\mathbf{r}|^{n+2}$, so:

$$= \kappa x_4^2 \left(\frac{\delta_{\eta\nu}}{|\mathbf{r}|^3} - \frac{3 x_\nu x_\eta}{|\mathbf{r}|^5} \right)$$

Differentiating for the $\partial \alpha_\nu / \partial x_4$ case:

$$\frac{\partial \alpha_\nu}{\partial x_4} = \frac{\kappa x_\nu}{|\mathbf{r}|^3} \frac{\partial}{\partial x_4} (x_4^2) = \frac{2 \kappa x_\nu x_4}{|\mathbf{r}|^3}$$

Differentiate once more for spatial indices η and ρ :

$$\frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_\eta} = \kappa x_4^2 \frac{\partial}{\partial x_\rho} \left(\frac{\delta_{\eta\nu}}{|\mathbf{r}|^3} - \frac{3 x_\nu x_\eta}{|\mathbf{r}|^5} \right) = \kappa x_4^2 \left(\frac{\partial}{\partial x_\rho} \left(\frac{1}{|\mathbf{r}|^3} \right) \delta_{\eta\nu} - 3 \frac{\partial}{\partial x_\rho} \left(\frac{x_\nu x_\eta}{|\mathbf{r}|^5} \right) \right)$$

By the standard result for $1/|\mathbf{r}|^n$, the first term:

$$\frac{\partial}{\partial x_\rho} \left(\frac{1}{|\mathbf{r}|^3} \right) = -\frac{3 x_\rho}{|\mathbf{r}|^5}$$

By product rule and the standard result for $1/|\mathbf{r}|^n$, the second term:

$$\frac{\partial}{\partial x_\rho} \left(\frac{x_\nu x_\eta}{|\mathbf{r}|^5} \right) = \frac{\partial x_\nu}{\partial x_\rho} x_\eta \frac{1}{|\mathbf{r}|^5} + x_\nu \frac{\partial x_\eta}{\partial x_\rho} \frac{1}{|\mathbf{r}|^5} + x_\nu x_\eta \frac{\partial}{\partial x_\rho} \left(\frac{1}{|\mathbf{r}|^5} \right)$$

$$= \frac{\delta_{\rho\nu}x_\eta}{|\mathbf{r}|^5} + \frac{\delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} - \frac{5x_\nu x_\eta x_\rho}{|\mathbf{r}|^7} = \frac{\delta_{\rho\nu}x_\eta + \delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} - \frac{5x_\nu x_\eta x_\rho}{|\mathbf{r}|^7}$$

Substituting in:

$$\begin{aligned} \frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_\eta} &= \kappa x_4^2 \left(-\frac{3x_\rho \delta_{\eta\nu}}{|\mathbf{r}|^5} - \frac{3\delta_{\rho\nu}x_\eta + 3\delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7} \right) \\ &= \kappa x_4^2 \left(\frac{-3x_\rho \delta_{\eta\nu} - 3\delta_{\rho\nu}x_\eta - 3\delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7} \right) \end{aligned}$$

For the $\partial^2 \alpha_\nu / \partial x_\rho \partial x_4$ case:

$$\begin{aligned} \frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_4} &= 2\kappa x_4 \frac{\partial}{\partial x_\rho} \left(\frac{x_\nu}{|\mathbf{r}|^3} \right) = 2\kappa x_4 \left(\frac{\partial x_\nu}{\partial x_\rho} \frac{1}{|\mathbf{r}|^3} + x_\nu \frac{\partial}{\partial x_\rho} \left(\frac{1}{|\mathbf{r}|^3} \right) \right) \\ &= 2\kappa x_4 \left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5} \right) \end{aligned}$$

by the standard result for $1/|\mathbf{r}|^n$. Note that the symmetry of mixed partials gives:

$$\frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_4} = \frac{\partial^2 \alpha_\nu}{\partial x_4 \partial x_\rho} = 2\kappa x_4 \left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5} \right)$$

Finally, the $\partial^2 \alpha_\nu / \partial x_4^2$ case:

$$\frac{\partial^2 \alpha_\nu}{\partial x_4^2} = \frac{2\kappa x_\nu}{|\mathbf{r}|^3} \frac{\partial x_4}{\partial x_4} = \frac{2\kappa x_\nu}{|\mathbf{r}|^3}$$

We can now substitute these blocks into the linearized Lavelle deflection equation:

$$\begin{aligned} \ddot{x}_\nu &= -\epsilon \frac{\partial^2 \alpha_\nu}{\partial x_k \partial x_l} \dot{x}^k \dot{x}^l = -\epsilon \left(\frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_\eta} \dot{x}^\rho \dot{x}^\eta + 2 \frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_4} \dot{x}^\rho \dot{x}^4 + \frac{\partial^2 \alpha_\nu}{\partial x_4^2} \dot{x}^4 \dot{x}^4 \right) \\ &= -\epsilon \left(+\kappa x_4^2 \left(\frac{-3x_\rho \delta_{\eta\nu} - 3\delta_{\rho\nu}x_\eta - 3\delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7} \right) \dot{x}^\rho \dot{x}^\eta \right. \\ &\quad \left. + 4\kappa x_4 \left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5} \right) \dot{x}^\rho \dot{x}^4 + \frac{2\kappa x_\nu}{|\mathbf{r}|^3} \dot{x}^4 \dot{x}^4 \right) \\ &\quad - \epsilon \kappa x_4^2 \left(\frac{-3x_\rho \dot{x}^\rho \dot{x}_\nu - 3x_\eta \dot{x}^\eta \dot{x}_\nu - 3x_\nu \dot{x}^\eta \dot{x}_\eta}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho \dot{x}^\rho \dot{x}^\eta}{|\mathbf{r}|^7} \right) \\ &= -4\epsilon \kappa x_4 \left(\frac{\dot{x}_\nu \dot{x}_4}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho \dot{x}^\rho \dot{x}_4}{|\mathbf{r}|^3} \right) - \frac{2\epsilon \kappa x_\nu}{|\mathbf{r}|^3} \dot{x}_4^2 \end{aligned}$$

where I've moved all the fours down to make things simpler. Grouping by denominator and cleaning up:

$$\ddot{x}_\nu = -\frac{2\epsilon \kappa x_\nu}{|\mathbf{r}|^3} \dot{x}_4^2 - \frac{4\epsilon \kappa x_4 \dot{x}_4}{|\mathbf{r}|^3} (\dot{x}_\nu - 3\dot{x}^\rho x_\rho x_\nu) + \frac{3\epsilon \kappa x_4^2}{|\mathbf{r}|^5} \left(-\dot{x}^\eta \dot{x}_\eta x_\nu - 2x_\rho \dot{x}^\rho \dot{x}_\nu + \frac{5\dot{x}^\eta x_\eta \dot{x}^\rho x_\rho x_\nu}{|\mathbf{r}|^2} \right)$$

3.2 Analyzing Gravity

The above equation simplifies dramatically for a spatially stationary particle, where $\dot{x}_\nu = 0, \dot{x}_\rho = 0, \dot{x}_\eta = 0$:

$$\ddot{x}_\nu = -\frac{2\epsilon\kappa x_\nu}{|\mathbf{r}|^3} \dot{x}_4^2$$

Rewriting $x_\nu/|\mathbf{r}|$ as $\hat{x}_\nu$:

$$\ddot{x}_\nu = -\frac{2\epsilon\kappa\hat{x}_\nu}{|\mathbf{r}|^2} \dot{x}_4^2$$

Identify $\epsilon = G/c^2$ and $\kappa = M/2$:

$$\ddot{x}_\nu = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2} \frac{\dot{x}_4^2}{c^2}$$

Later, we'll find out why this matches Newtonian gravity exactly for a stationary particle.

For a moving particle, corrections are picked up; the first of which are:

$$-\frac{4GMx_4\dot{x}_4\dot{x}_\nu}{2c^2|\mathbf{r}|^3} + \frac{12GMx_4x_\nu\dot{x}_4\dot{x}^\rho x_\rho}{2c^2|\mathbf{r}|^3}$$

The first term scales linearly with velocity and points in the $-\dot{x}_\nu$ direction, creating a drag on velocity. The second term scales with the dot product of $\dot{\mathbf{x}}$ and $\mathbf{x}$ and points in the x_ν direction, creating a radial boost that activates for velocity. In the early-time limit (small x_4), these terms vanish as well. In the late-time limit, they dominate. The second set of corrections are:

$$-\frac{3GMx_4^2\dot{x}^\eta\dot{x}_\eta x_\nu}{2c^2|\mathbf{r}|^5} - \frac{3GMx_4^2x_\rho\dot{x}^\rho\dot{x}_\nu}{c^2|\mathbf{r}|^5} + \frac{15GMx_4^2\dot{x}^\eta x_\eta\dot{x}^\rho x_\rho x_\nu}{2c^2|\mathbf{r}|^7}$$

The first term scales with the squared magnitude of velocity, pointing in the x_ν direction, pushing away from the origin. The second term is linear in the dot of position and velocity, so it activates for radial motion and drags velocities. The final term is quadratic in the dot of position and velocity, pushing away from the origin. These all vanish in the early-time limit.

Thus, based on our analysis we can say that Newtonian gravity lives in the early-time and low-velocity limit of the plastica.

A small extra implication of gravity SPT includes is length expansion, which comes from Lavelle's equation:

$$dx_\nu = \Lambda_{\nu i} dC^i = \left(1 - \epsilon \frac{\partial \alpha_\nu}{\partial x_i}\right) dC^i = dC^i \left(1 - \epsilon \kappa x_4^2 \left(\frac{\delta_{i\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_i}{|\mathbf{r}|^5}\right) - \frac{2\epsilon \kappa x_\nu x_4}{|\mathbf{r}|^3}\right)$$

Set $\epsilon = G/c^2$ and $\kappa = M/2$:

$$= dC^i \left(1 - \frac{GMx_\nu x_4}{c^2|\mathbf{r}|^3} - \frac{GMx_4^2}{2c^2|\mathbf{r}|^3} \left(\delta_{i\nu} - \frac{3x_\nu x_i}{|\mathbf{r}|^2}\right)\right)$$

Lengths contract according to the equation above in the plastica. In the early-time limit, $x_4 > x_4^2$ and the first correction dominates.

3.3 Normal Kinematics

We'll be using a familiar constraint to pin α_4 called the “normal” kinematics (NK) constraint:

$$|\dot{\mathbf{x}}| = c \text{ always}$$

This constraint allows phenomena we see in “normal” space, or a nearly flat plastica. For a stationary particle, $\dot{\mathbf{r}} = \mathbf{0}$, so $\dot{x}_4 = c$. Plugging into our stationary particle Lavelle deflection equation:

$$\ddot{x}_\nu = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2} \frac{\dot{x}_4^2}{c^2} = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2} \frac{c^2}{c^2} = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2}$$

We get Newtonian gravity exactly.

Rewriting this constraint with a dot product and differentiating once:

$$\begin{aligned} \dot{x}_k \dot{x}^k = c^2 &\implies \frac{dc}{dC_4} = 0 = \frac{d}{dC_4} (\dot{x}_k \dot{x}^k) = \ddot{x}^k \dot{x}_k + \dot{x}^k \ddot{x}_k = 2\ddot{x}^k \dot{x}_k = 0 \\ &\implies \ddot{x}^k \dot{x}_k = 0 \end{aligned}$$

Acceleration must be purely tangential to the 4-velocity. Splitting into temporal and spatial parts:

$$\ddot{x}^\nu \dot{x}_\nu + \ddot{x}_4 \dot{x}_4 = 0 \implies \ddot{x}^\nu \dot{x}_\nu = -\ddot{x}_4 \dot{x}_4$$

where ν runs over spatial indices. This pins α_4 ; the definition is:

$$\frac{\ddot{x}^\nu \dot{x}_\nu}{\dot{x}_4} = \epsilon \frac{\partial^2 \alpha_4}{\partial x_k \partial x_l} \dot{x}^k \dot{x}^l$$

Substituting in $\ddot{x}_\nu$ and raising $\dot{x}^\nu$ to match:

$$\begin{aligned} \epsilon \frac{\partial^2 \alpha_4}{\partial x_k \partial x_l} \dot{x}^k \dot{x}^l = & -\frac{2\epsilon\kappa x_\nu \dot{x}^\nu}{|\mathbf{r}|^3} \dot{x}_4 - \frac{4\epsilon\kappa x_4 \dot{x}^\nu}{|\mathbf{r}|^3} (\dot{x}_\nu - 3\dot{x}^\rho x_\rho x_\nu) \\ & + \frac{3\epsilon\kappa x_4^2 \dot{x}^\nu}{|\mathbf{r}|^5 \dot{x}_4} \left(-\dot{x}^\eta \dot{x}_\eta x_\nu - 2x_\rho \dot{x}^\rho \dot{x}_\nu + \frac{5\dot{x}^\eta x_\eta \dot{x}^\rho x_\rho x_\nu}{|\mathbf{r}|^2} \right) \end{aligned}$$

Notice the $\dot{x}_4$ in the denominator of the last term in the LHS, and the cubic in velocity in the same term. This implies that α_4 itself carries a velocity dependent structure. Also notice the signature; the LHS is directly the second derivative, whereas in other components it is the negative second derivative, which is an interesting appearance of a Lorentzian signature from the NK constraint.

Notice that this can only be done because we have an extra degree of freedom in α^4 ; should it have been determined by the Lagrangian, we wouldn't have been able to add another equation to the system.

Conclusions

Throughout this paper, we've investigated the implications of a plastica-based universe and explored the outcomes thoroughly.

Beginning with space as a deformable solid medium, we derived the conversions of positions, velocities, and accelerations between our two coordinate systems. We also derived the sourcing equation of the plastica from a Lagrangian action and used it to solve for the general sourcing equation for any continuous mass; we then created a gravitational perturbation and constrained masses moving through the plastica to recover Newtonian gravity exactly.

There are still, however, some open questions:

- Does the plastica admit gravitational waves?
- Can SPT reproduce General Relativity under the correct choice of action?
- What does SPT look like at second order/higher orders?
- Does light in SPT behave correctly?

These questions are still under research, and their answers yet to be determined.